

Session 4 homework: filters, Gaussian blur and Canny on MILK10k

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Notebook (code and all outputs): https://github.com/sebarios02/Milk10k-project/blob/main/notebooks/session4_homework_filters_canny.ipynb

Data: MILK10k dermoscopic images only (same device type for every image). Paths come from `milk10k_pipeline/config.py`; nothing is hard-coded. All filters run on the grayscale image scaled to $[0, 1]$.

1. Explore filters across diagnosis classes

Apply a set of filters to MILK10k images from different diagnosis classes. Do edges or textures look visually different between malignant and benign lesions?

Method. One dermoscopic image from each of six classes (malignant: invasive melanoma, basal cell carcinoma, squamous cell carcinoma; benign: nevus, seborrheic keratosis, dermatofibroma), each picked from a seeded random sample. Five filters seen in class: Gaussian blur ($\sigma = 3$), 3×3 sharpen, Sobel gradient magnitude (after $\sigma = 1$ smoothing) for **edges**, $|\text{Laplacian of Gaussian}|$ ($\sigma = 2$) for dots and blobs, and the 9×9 local standard deviation as a simple **texture** measure. Each filter column uses one common colour scale for all six images, so a brighter panel really means a stronger response.

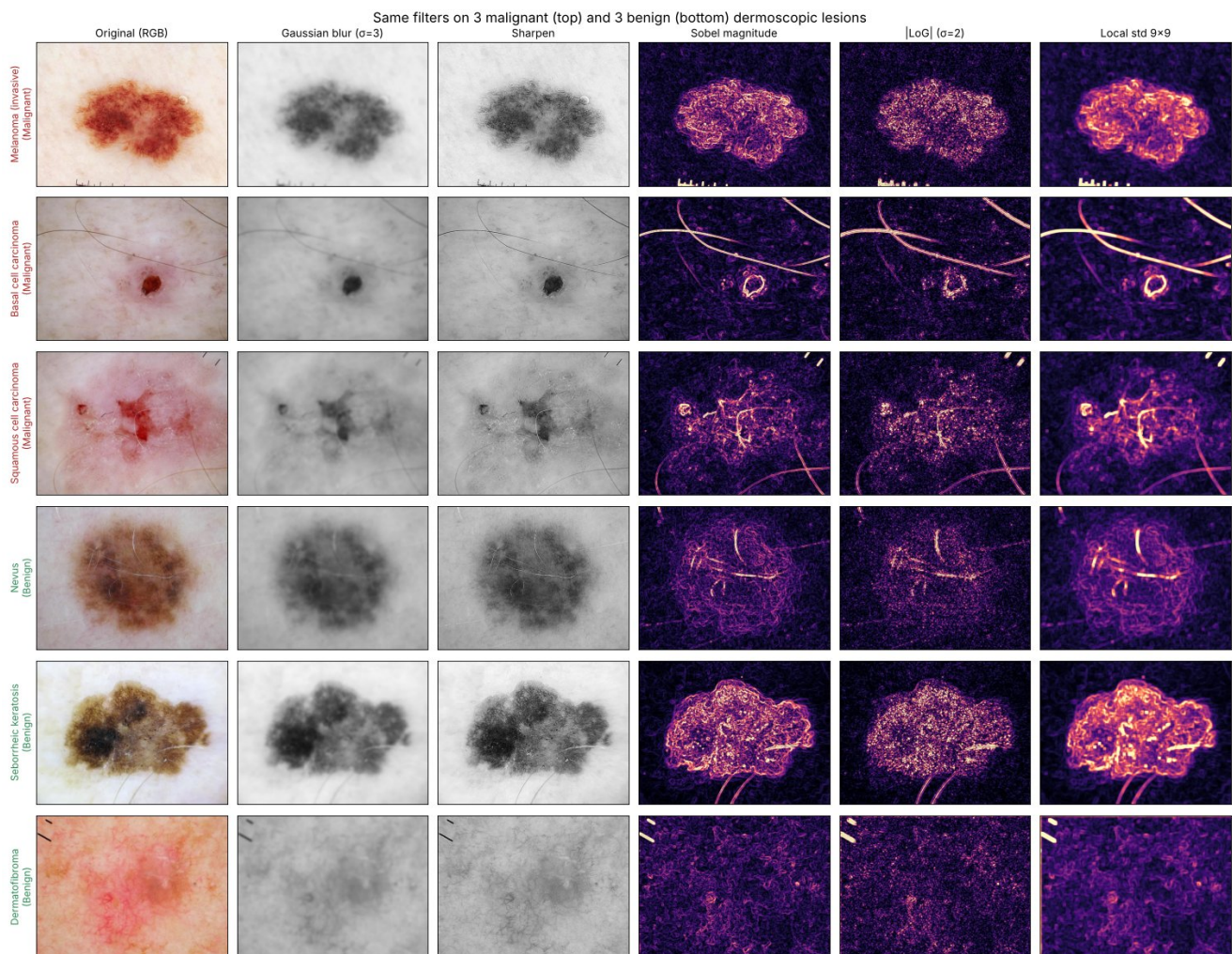


Figure 1. Same filters on 3 malignant (top rows) and 3 benign (bottom rows) dermoscopic lesions.

What the examples show.

- **Edges are dominated by things that are not the lesion.** On the BCC, the SCC and the nevus, the strongest Sobel, |LoG| and local-std responses are the **hairs**, not the lesion border. Ruler ticks and pen marks are as bright as anything inside the lesion.
- **Texture is high in both groups.** The melanoma (malignant) and the seborrheic keratosis (benign) have the busiest interiors of all six (pigment network, dots, crypts). Strong internal texture belongs to pigmented, structured lesions in general, not to malignancy.
- **Borders differ in sharpness, but not along benign vs malignant.** The melanoma has a thick but irregular outline that fades unevenly into the skin, the nevus outline is weak and hidden behind hairs, the seborrheic keratosis has the sharpest "stuck-on" border, and the dermatofibroma has almost no edge.
- **Grayscale removes the main cue of several malignant classes.** The SCC and BCC are recognisable by red or pink areas and vessels, which become low-contrast patches in grayscale.

Check on a random sample. To avoid cherry-picking, I computed three numbers on 40 benign and 40 malignant random dermoscopic images (seed 42): texture inside the lesion (mean local std), border strength (mean Sobel magnitude on a thin ring around the lesion) and whole-image edge strength. The lesion mask is a rough Otsu threshold on the blurred image (the largest dark region not touching the frame). Groups were compared with a Mann-Whitney U test.

Feature	Median benign	Median malignant	Mann-Whitney p
Texture inside lesion (local std)	0.028	0.023	0.19
Border strength (Sobel on border ring)	0.100	0.064	0.034
Whole-image edges (Sobel, all pixels)	0.052	0.046	0.59

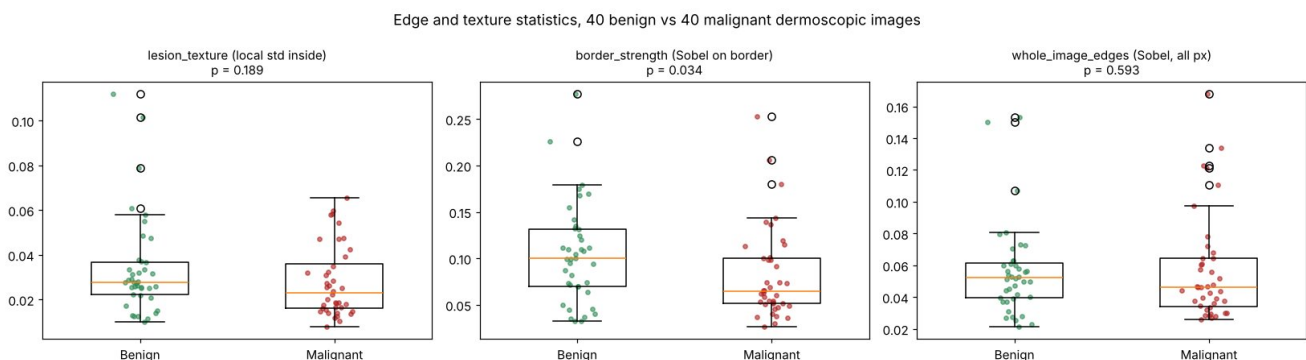


Figure 2. Edge and texture statistics. Sample composition: benign = 18 nevi, 10 seborrheic keratoses, 12 other; malignant = 28 BCC, 8 melanomas, 4 SCC/keratoacanthoma.

Answer. In individual images, edges and textures do look different from lesion to lesion, but **not in a way that separates benign from malignant**. On the random sample the distributions overlap heavily. The only borderline difference goes the "wrong" way: *benign* lesions have the stronger border ($p \approx 0.03$, not significant after a Bonferroni correction for three tests, $0.03 \times 3 \approx 0.10$). This is explained by the class mix: most benign images are dark pigmented nevi and keratoses with high contrast to the skin, while 28 of the 40 malignant images are BCCs, often pale pink and barely darker than the skin in grayscale. For the same reason, the median lesion mask covers 10% of a benign image but only 0.7% of a malignant one. Hair, rulers and the loss of colour are the main confounders. A single hand-crafted filter is not a classifier, which is exactly why a CNN learns many filters, on colour images, and combines them.

2. Gaussian blur: what does sigma control?

Read the `scipy.ndimage.gaussian_filter` documentation and identify what the `sigma` parameter controls. Experiment with different values of `sigma` and observe the effect.

```
gaussian_filter(input, sigma, order=0, mode='reflect', cval=0.0, truncate=4.0, radius=None, axes=None)
```

- **sigma is the standard deviation of the Gaussian kernel, in pixels.** It sets the width of the bell curve, i.e. how far a neighbour can be and still contribute to the weighted average. Small σ means light blur; large σ means strong blur over a wide neighbourhood.
- It can be a single number (same blur on every axis) or one value per axis, e.g. `sigma=(0, 5)` blurs only horizontally.
- The kernel is cut at `truncate × sigma` (default 4σ), so its radius is `int(4 $\sigma$  + 0.5)`: 4 px for $\sigma = 1$, 16 px for $\sigma = 4$, 64 px for $\sigma = 16$. A larger σ is also slower.
- The weights sum to 1, so blurring keeps the average brightness and only spreads it. In frequency terms it is a low-pass filter: detail smaller than about σ pixels disappears first. With `order > 0` it returns derivatives of the Gaussian instead (used for edge detection).

ISIC_2713587 (nevus): Gaussian blur with growing σ (top) and the edges that survive (bottom)

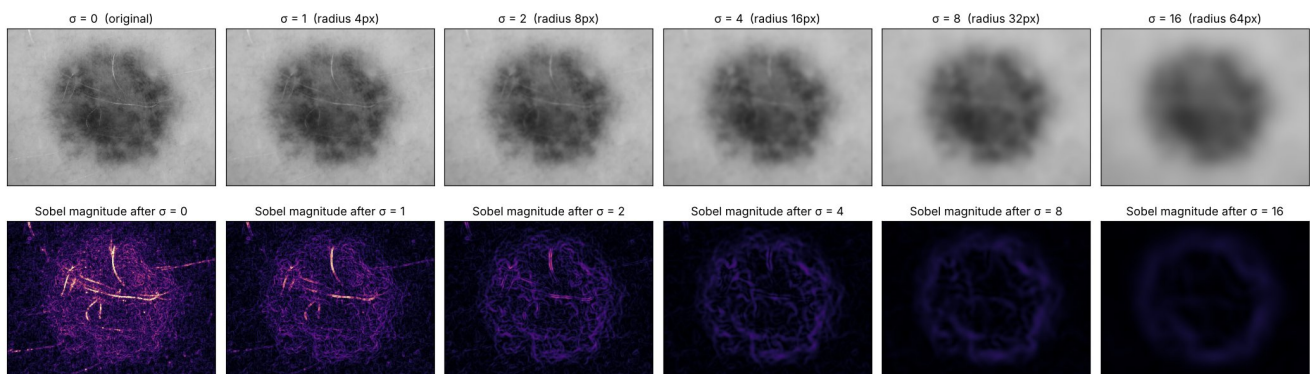


Figure 3. Nevus with hairs (ISIC_2713587). Top: blur with growing σ . Bottom: Sobel magnitude of each blurred image (same colour scale), i.e. the edges that survive.

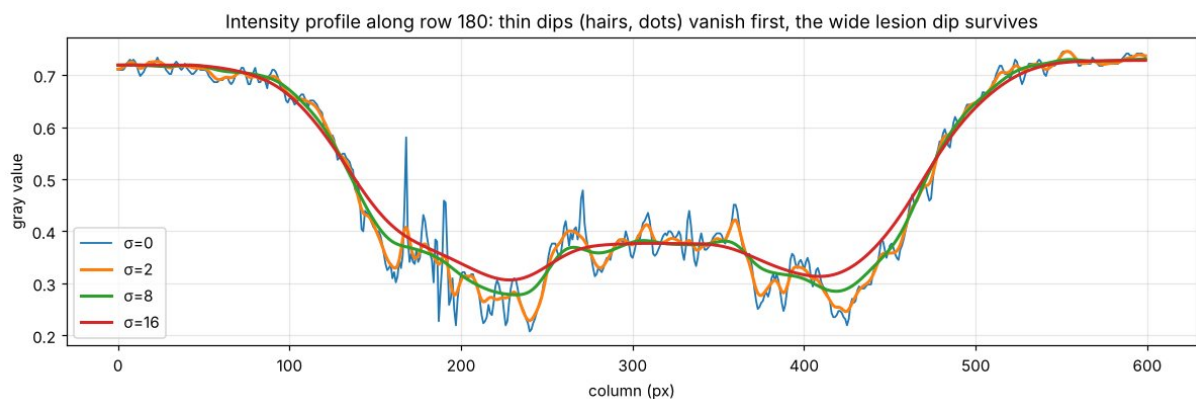


Figure 4. Intensity along one row through the lesion. Hair spikes and pigment dots vanish at small σ ; the wide lesion "valley" survives even at $\sigma = 16$, with softer walls.

σ	0	1	2	4	8	16
Kernel radius (px)	–	4	8	16	32	64
Mean brightness	0.581	0.581	0.581	0.581	0.581	0.581
Mean Sobel magnitude	0.073	0.051	0.035	0.024	0.019	0.015

Observations. $\sigma = 1$ removes JPEG noise and the finest dots. $\sigma = 2-4$ is the useful range for these 600×450 images: thin hairs fade and their edge response almost disappears, while the lesion outline is still clear. $\sigma = 8-16$ blurs the lesion itself, so the border becomes a soft ramp and only a faint ring of edges is left. Mean brightness never changes, but the mean edge strength falls by a

factor of five between $\sigma = 0$ and $\sigma = 16$. **σ is a scale knob**: it decides the size of the details you keep. A small σ before edge detection is a cheap noise filter for MILK10k, but no σ removes hair without also softening the lesion border, because hairs and fine pigment structures live at the same scale.

3. Canny edge detection, implemented from scratch

Implement Canny as a Python function, apply it to one MILK10k image and compare it with a Sobel edge detector.

Implementation (function `canny(gray, sigma, low, high)` in the notebook, NumPy + `scipy.ndimage` only):

- **Gaussian smoothing** with σ to reduce noise.
- **Gradient** with Sobel kernels: magnitude $\sqrt{G_x^2 + G_y^2}$ and direction $\text{atan2}(G_y, G_x)$.
- **Non-maximum suppression**: the direction is rounded to 0/45/90/135°, and a pixel is kept only if it is at least as strong as its two neighbours across the edge. Thick ridges become 1-pixel lines.
- **Double threshold**: above *high* = strong edge, between *low* and *high* = weak edge (both relative to the maximum gradient; here *low* = 0.08, *high* = 0.20).
- **Hysteresis**: connected components (8-neighbourhood) of strong + weak pixels are kept only if they contain at least one strong pixel, so faint parts of a contour survive and isolated noise is dropped.

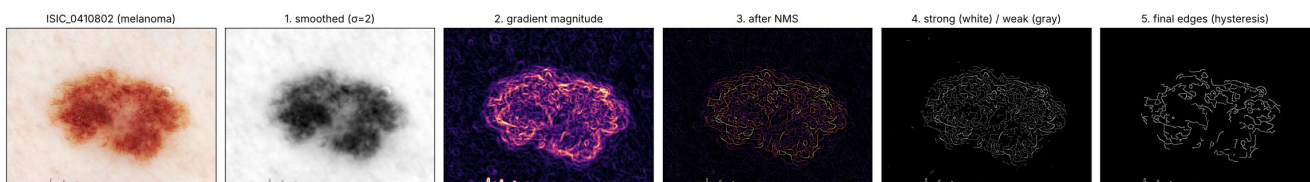


Figure 5. The five Canny steps on an invasive melanoma (ISIC_0410802).

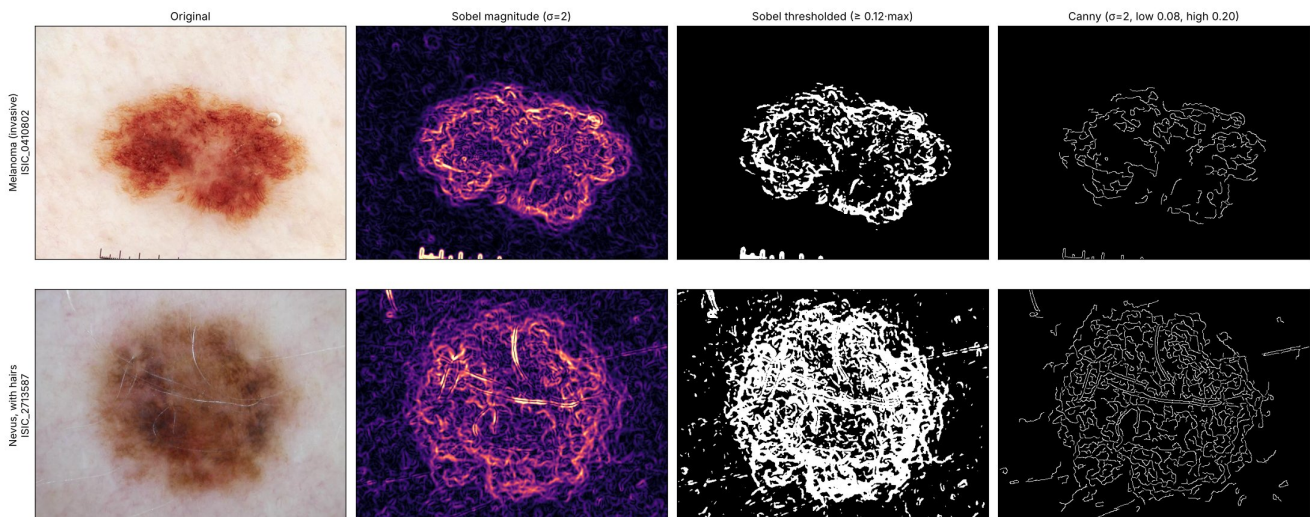


Figure 6. Sobel vs Canny with the same $\sigma = 2$ pre-smoothing. The Sobel edge map uses one threshold ($0.12 \times \text{max}$, between Canny's two thresholds). Second row: a nevus with hairs, to show behaviour on hair.

Image	Detector	% edge pixels	Connected segments	Approx. line width (px)
Melanoma	Sobel thresholded	9.4	224	3.7
Melanoma	Canny	2.0	122	1.0
Nevus with hairs	Sobel thresholded	32.1	427	4.3
Nevus with hairs	Canny	6.1	524	1.0

Line width $\approx$ edge pixels $\div$ edge pixels that are ridge maxima (NMS). Canny is 1 px by construction.



Figure 7. Zoom on the melanoma border.

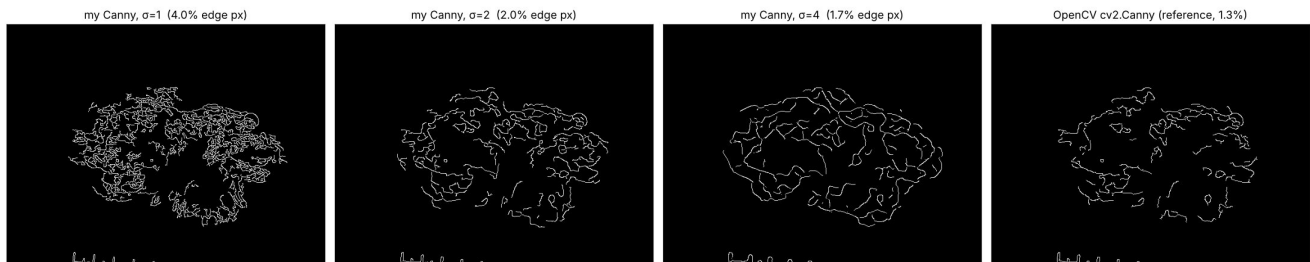


Figure 8. My Canny with $\sigma = 1, 2, 4$, and OpenCV's cv2.Canny as a reference.

Main differences observed:

- **Thin vs thick.** Sobel responds on both sides of a boundary, so its thresholded edges are bands of about 4 px. Canny's non-maximum suppression keeps only the ridge top: 1-px, better-localised edges (clearest in Figure 7).
- **Thresholding.** Sobel gives a continuous strength map, and one threshold is a bad compromise. On the nevus the same rule marks a third of the image as "edge", because hairs set the maximum and skin texture passes the threshold. Canny's double threshold with hysteresis keeps the output sparse (2–6% of pixels).
- **Continuity.** Hysteresis follows faint contour stretches attached to strong ones, giving fewer, longer curves on the melanoma (122 vs 224 segments). On the nevus Canny has more segments only because Sobel's thick bands merge hair, skin texture and lesion into a few huge blobs.
- **Explicit scale.** With $\sigma = 1$ Canny traces the pigment network inside the melanoma, with $\sigma = 4$ mainly the large structures and the outline (Figure 8). Plain 3×3 Sobel has a fixed, very small scale.
- **Shared weaknesses.** Both fire on hairs (Canny turns each hair into two parallel lines), on the ruler ticks and on the air bubble. On the melanoma, Canny at $\sigma = 2$ finds internal pigment boundaries more than the outer border, because the border fades gradually and its gradient is weaker than the internal contrasts.
- **Cost.** Canny has 5 steps and 3 parameters (σ , low, high) and is slower; Sobel is two small convolutions. My implementation gives edge maps close to OpenCV's cv2.Canny, which supports that the steps are correct.

Conclusion

Hand-crafted filters show real differences between individual lesions, but they respond to pigmentation, hair and artefacts at least as much as to malignancy, and on a random sample they do not separate benign from malignant. σ controls the scale of what a Gaussian removes; small σ is a useful pre-processing step, while large σ destroys the lesion border. Canny turns the Sobel gradient into thin, connected, sparse edges, but neither detector can tell a lesion border from a hair. These limits are the motivation for learned filters (CNNs) trained on colour images.